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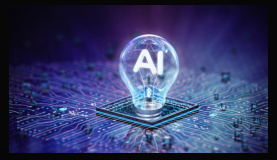
AI Fearlessly



Defining AI Stack Telemetry

Detection, Incident Response, Compliance, Governance

Global Digital Foundation, March 4th 2026, Barcelona, Spain

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The AI Observability Crisis

Case Study

MS 365 Copilot Agent Exploit - EchoLeak

Incident
Researchers demonstrated a prompt injection attack via email with no Human interventions aka 0 Click

Impact
Sensitive data exfiltration

Missing Observability
No user-facing audit trail, no visibility into data access, opaque tool execution, limited security telemetry, no decision explainability

Outcome
Demonstrated the need for security visibility onto agent behavior

Why Now: The Perfect Storm

Rapid AI Adoption
Enterprise AI Deployments accelerating

Regulatory Pressure
EU AI Act, emerging compliance requirements

Visibility Gap
Traditional monitoring blind to AI behavior

Security Incidents
Growing Attack Surface, novel threats

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Why Now? The Convergence of Pressures

Four forces demanding comprehensive AI telemetry

Regulatory Mandates

- EU AI ACT
- NIST AI RMF
- ISO 42001

- Earlier Privacy, Vertical or National regulations
  - SOC2
  - ISO 27001
  - PCI DSS 4.0
  - GDPR

Security Threats

- MITRE ATLAS
  - 66+ techniques
  - 14 new agentic

- OWASP
  - Top 10 LLM
  - Top Agentic

Production Reality

- Experiments → Mission critical
- Customer facing
- Revenue impacting
- Troubleshouting
- SecOps
- Risk Management

Agentic Future

- **Autonomous agents**
- **Tool calling**
- **Multi-agents systems**
- **Agentic Identity proliferation**

The OpenClaw Eye Opening

These are just a few examples

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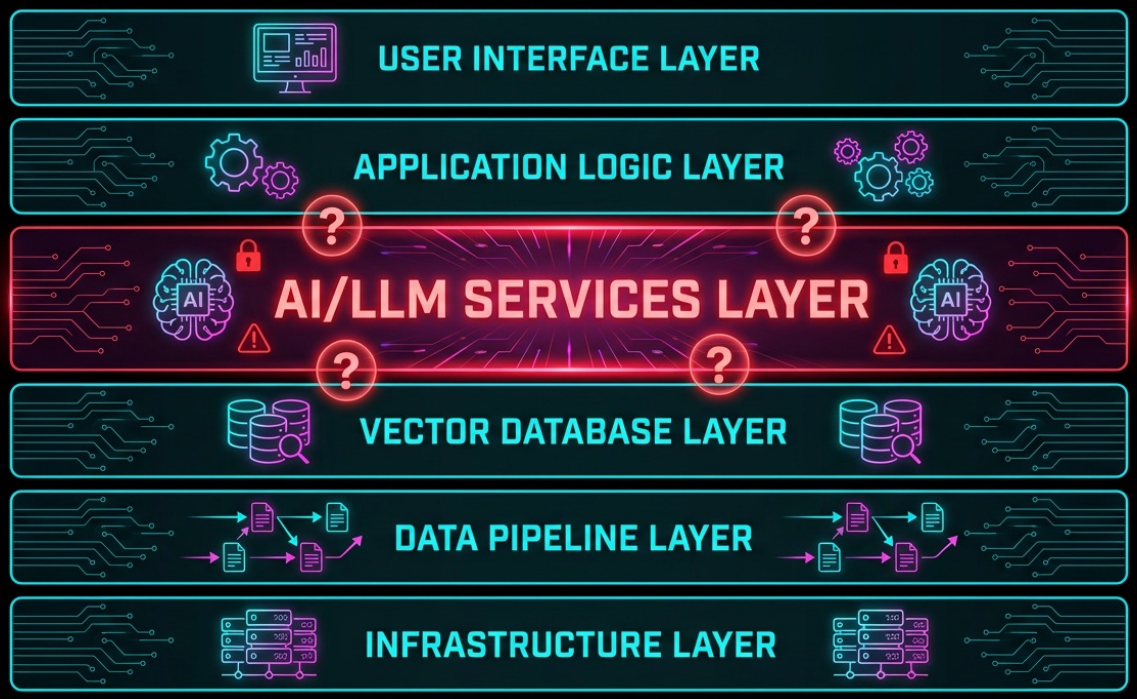
The AI Stack Visibility Gap

Traditional monitoring tools miss critical AI specific layers



OSS Vector DB like FAISS has no logging at all.

Gap between User/Free & Enterprise Editions

Some AI Services provide basic logs with low visibility into tool access or reasoning.

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Organizational Barriers

The Trilemma

Cybersecurity says:

"Log Everything"

You can't easily build behavioral baselines, run anomaly detection, or do incident forensics without the actual content.

Privacy says:

Log nothing (or as little as possible):

Under GDPR (Art. 5 data minimization), and PIPEDA (Canada), collecting and storing prompt content creates a new data processing activity that requires justification, consent mechanisms, and retention policies.

Compliance says:

"Log selectively but prove you did it right."

Regulations like SOC 2, ISO 27001, HIPAA, and industry-specific frameworks require audit trails — but also require you to demonstrate you're not over-collecting. PCI DSS 4.0, for instance, would flag prompt logs containing credit card numbers as a scoping issue.

5 Barriers to AI Logging

1. **Privacy Concerns**
   Fear of logging sensitive user data
2. **Performance Impact**
   Perceived latency from logging
3. **Cost Considerations**
   Storage and processing overhead & Token Economics
4. **Technical Complexity**
   No standardized AI telemetry format & Multi-step reasoning
5. **Organizational Silos**
   ML teams disconnected from SecOps

Be Creative: Decoupling detection from storage and **separating sensitivity tiers:**

-**Security** gets what it needs: real-time detection at the endpoint plus metadata for hunting/correlation at scale

–**Privacy** gets data minimization by default: full prompts never leave the local environment unless there's a justified investigation. Encryption + CMK

-**Compliance** gets defensible architecture: you can demonstrate purpose limitation, minimization, and access controls to auditors

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Framework Landscape Overview

Analyzing AI security frameworks for telemetry requirements

Framework	Focus Area	Telemetry Depth	Maturity
OWASP LLM Top 10	Application vulnerabilities	**Medium**	Established
OWASP Agentic Security	Agent-specific risks	**High**	Emerging
MITRE ATLAS	Adversarial ML tactics	**High**	Established
CSA AI Controls Matrix	Cloud AI governance	**Medium**	Emerging
NIST AI RMF	Risk management	**Low**	Established
ISO 42001	AI management systems	**Low**	Established
OpenTelemetry GenAI SIG	Gen AI Telemetry	**Low**	Low

Note: Partial list of frameworks

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OWASP LLM & Agentic + MITRE ATLAS

OWASP Top 10 LLM & Agentic

Key Risks:

Top 10 LLM

- Prompt Injection
- Sensitive Information Disclosure
- Excessive Agency
- Supply Chain Vulnerabilities

Top 10 Agentic

- Excessive Agency
- Tool Misuse
- Unsafe Delegation

Telemetry Requirements:

- Input/output logging for injection detection
- Data flow monitoring for PII exposure
- Agent action audit trails
- Dependency and model provenance tracking

Coverage: Application & Agent Layer

MITRE ATLAS

Key Tactics:

- Reconnaissance
- Initial Access
- ML Attack Staging
- Exfiltration

Telemetry Requirements:

- Model query patterns and rate limiting
- Training data access logs
- Model behavior drift detection
- Adversarial input signatures

Coverage: Adversarial ML Lifecycle

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ISO 42001 + Cross-Framework Telemetry Mapping

ISO 42001
AI Management Systems

Key Clauses:

- 6.1: Risk Assessment
- 7.5: Documented Information
- 8.4: AI System Development
- 9.1: Monitoring & Measurement

Telemetry Implication:

Requires evidence of AI system monitoring for certification

Telemetry Category	OWASP	MITRE	CSA	NIST	ISO
Input/Output Logging	●	●	●	●	●
Model Performance	●	●	●	●	●
Data Lineage	●	●	●	●	●
Security Events	●	●	●	●	●
Agent Actions	●	●	●	●	●

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Introducing AITF

Github

AI Telemetry Framework — Bridging Observability and Security

OpenTelemetry GenAI

Traces, Metrics & Events
gen_ai.* semantic conventions
Model, Agent, Tool spans

OCSF Security Schema

Normalized Security Events
New Category 7: AI/ML Activity
SIEM/XDR native ingestion

AITF Architecture

L1 Instrumentation	OTel GenAI SDK + AITF Extensions
L2 Collection	OTel Collector + Security Processor
L3 Normalization	AITF OCSF Mapper
L4 Analytics	XDR / SIEM / Data Analytics

One or Dual pipeline. Full coverage from developer to SOC.

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AI Telemetry Framework (AITF)

Available soon

PROPOSED FRAMEWORK

- **Comprehensive telemetry framework** for AI systems
- **Extends OTel GenAI SIG** — not a replacement
- **Bridges observability & security** via dual pipeline
- **15 semantic namespaces** (agent, MCP, skill, RAG, identity…)
- **10 OCSF event classes** for SIEM/XDR integration
- **Proposed to CoSAI** Working Stream 2

GOALS

- **Standardise AI telemetry** across the industry
- **First-class agentic support** — MCP, skills, multi-agent
- **Compliance by design** — 8 frameworks, 305 controls
- **Real-time threat detection** — 14 built-in rules
- **Production ready** — stable conventions, 3 SDKs
- **OWASP alignment** — LLM, Agentic & MCP Top 10

TECHNOLOGY

- **4-layer pipeline:** Instrument → Collect → Normalise → Analyse
- **OTel Collector processors:** Security, PII, Cost, Compliance
- **OCSF Category 7 mapper** for security event normalisation
- **SDKs:** Python, Go, TypeScript (Java/C++ roadmap)
- **Detection engine:** Sigma rules, Splunk SPL, anomaly scoring
- **Identity:** OAuth 2.1, SPIFFE, mTLS, delegation chains

BENEFITS

- **Full agent lifecycle visibility** — reasoning to delegation
- **Threat detection at AI layer** — injection, exfiltration, hijack
- **Cost attribution** — per-request, per-user, per-model
- **Supply chain security** — provenance, AI-BOM, integrity
- **Memory security** — poisoning detection, isolation
- **SIEM integration** — AWS Security Lake, Splunk, TrendAI Vision One…

AITF Essential Categories

Building on OpenTelemetry GenAI SIG with comprehensive AI observability

CAPABILITY	OTel	AITF	AITF DETAIL	PRIO
LLM Inference Tracing	Experimental	AITF Coverage	Stable + Enhanced	P1
Agent Spans	Experimental	AITF Coverage	Full lifecycle, delegation, memory	P1
MCP Protocol	Not Covered	AITF Coverage	Native aitf.mcp.* namespace	P1
Skills / Tool Registry	Not Covered	AITF Coverage	Full aitf.skill.* namespace	P1
Multi-Agent Orchestration	Not Covered	AITF Coverage	Teams, delegation, consensus	P2
Security Events	Not Covered	AITF Coverage	OWASP LLM Top 10, threat detection	P1
OCSF Integration	Not Covered	AITF Coverage	Native OCSF Category 7 export	P2
Compliance Mapping	Not Covered	AITF Coverage	8 frameworks (NIST, MITRE, EU AI Act...)	P2
Cost Attribution	Not Covered	AITF Coverage	Per-request, per-user, per-model	P2
Quality Metrics	Not Covered	AITF Coverage	Hallucination, confidence, factuality	P3
PII Detection	Not Covered	AITF Coverage	Built-in processor	P2
Supply Chain	Not Covered	AITF Coverage	Model provenance, AI-BOM	P3
Guardrail Telemetry	Not Covered	AITF Coverage	Content filtering, safety checks	P2
Agentic Identity	Not Covered	AITF Coverage	OAuth 2.1, SPIFFE, delegation chains	P1
LLMOps / MLOps	Not Covered	AITF Coverage	Training, eval, deploy, drift, prompts	P3

Experimental
AITF Coverage
Not Covered

Priority: P1 — Foundational (Phase 1) P2 — Core (Phase 2) P3 — Extended (Phase 3)

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AITF Event Classes

OCSF Category 7: AI/ML Activity — Eight classes covering the full AI attack surface

7001

AI Inference

Prompts, responses, token usage, model params

7002

AI Agent Activity

Orchestration, planning, delegation, boundaries

7003

AI Tool Execution

MCP skills, functions, API calls, permissions

7004

AI Data Retrieval

RAG pipeline, vector DB, knowledge base access

7005

AI Security Finding

Guardrails, content filters, policy engine results

7006 v1.1

AI Supply Chain

Model provenance, skill integrity, AI-BOM, signing

7007 v1.1

AI Integrity

Config drift, agent def tampering, guardrail bypass

7008 v1.1

AI Identity

Agent auth, credentials, delegation chains, trust

Natively ingestible by OCSF-compatible SIEMs

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What Makes AITF Different

Beyond inference monitoring — securing the entire AI lifecycle

AI Supply Chain Security

- Model hash verification
- Skill provenance tracking
- AI-BOM generation
- Registry trust scoring
- Dependency monitoring
- Signature verification

Integrity Monitoring

- Agent definition baselines
- Skill manifest mutation detect
- Guardrail policy drift alerts
- Config change approval flow
- Prompt template tracking
- Automated rollback support

Agentic Identity

- Agent authentication (mTLS)
- Delegation chain tracking
- Credential lifecycle mgmt
- Zero-trust for AI agents
- A2A identity federation
- Privilege escalation alerts

AITF doesn't just monitor inference — it secures the entire AI supply chain and identity layer.

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AITF for AI Forensics & Incident Response

Forensic-grade telemetry enabling full reconstruction of AI incidents across agent lifecycles, tool chains & identity paths

Agent Lifecycle Reconstruction

- Full session → step → action trace
- ReAct chain-of-thought reasoning log
- Scratchpad state at every decision point
- Turn count & loop detection evidence

Attack Chain Analysis

- MCP tool invocation with input/output
- Read-then-send exfiltration patterns
- 14 detection rules with MITRE ATLAS mapping
- Sigma rules for cross-SIEM deployment

Identity & Delegation Forensics

- Full delegation chain tracking & depth
- Scope attenuation verification
- OAuth 2.1 / SPIFFE / mTLS audit trail
- Session hijack & privilege escalation detection

Memory Poisoning Investigation

- Content hash & integrity verification
- Provenance tracking (trusted vs untrusted)
- Mutation count & poisoning anomaly score
- Cross-session isolation enforcement

Compliance Evidence & Audit

- 8 frameworks — NIST, MITRE, EU AI Act...
- 305 mapped controls, auto-generated evidence
- Tamper-evident logging (ISO 42001)
- OCSF export to SIEM/XDR platforms

Drift & Supply Chain Forensics

- Statistical drift tests with root cause
- Blast radius: isolated → segment → global
- Model provenance & cryptographic signing
- AI-BOM generation & dependency audit

OCSF Event Classes for Forensics:

7002: Agent Activity | 7003: Tool Execution | 7005: Security Finding | 7006: Supply Chain | 7007: Governance | 7008: Identity | 7010: Asset/Drift/Memory

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AITF Implementation Roadmap

PHASE 1 — Months 1–3

PHASE 2 — Months 4–6

PHASE 3 — Months 6–18

FOUNDATION

Core Specs & SDKs

Semantic Conventions

- gen_ai.* stable LLM tracing
- aitf.agent.* lifecycle & reasoning
- aitf.mcp.* protocol instrumentation
- aitf.skill.* tool registry

Core Infrastructure

- Python, Go, TypeScript SDKs
- OTel Collector base config
- Security processor (OWASP Top 10)
- PII detection & redaction processor

Identity & Auth

- aitf.identity.* — OAuth 2.1, SPIFFE
- Delegation chain tracking

CORE EXPANSION

Security, Compliance & Ops

Security & Detection

- 14 detection rules deployment
- Sigma rules for cross-SIEM
- OCSF Category 7 mapper (10 classes)
- Guardrail telemetry processor

Compliance & Cost

- 8-framework compliance mapping
- Cost attribution processor
- Per-request / per-user / per-model
- aitf.asset.* inventory & audit

Integrations

- SIEM export (Splunk, AWS Sec Lake, TrendAI)
- Adoption by CoSAI members and major vendors

SCALE & MATURE

Platform, Ecosystem & AI-BOM

Advanced Capabilities

- aitf.quality.* hallucination metrics
- aitf.drift.* detection & investigation
- aitf.supply_chain.* provenance, AI-BOM
- aitf.model_ops.* full MLOps lifecycle

Platform Expansion

- Java & C++ SDK releases
- Kubernetes Operator (auto-instrument)
- Native OTel Collector processor plugin
- Real-time Grafana threat dashboards
- AI-BOM generator from telemetry

Ecosystem

- Wide Industry Support

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"You can't secure what you can't see"

"AITF: Making AI visible to the defenders who need it most"

Security Axiom for the AI Era

AI telemetry is not optional—it's the foundation of AI security and governance.

Start Today

- Audit your AI stack visibility
- Prioritize high-value telemetry
- Build toward the standard

Questions?

Thank You

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AI Fearlessly

David Girard

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Support Slides

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Threat & Governance Mapping

Threat Detection

Interaction	Prompt injection, social engineering
Model	Model theft attempts, evasion attacks
Security	All OWASP LLM Top 10 threats
Data	Data exfiltration, poisoning
Operational	DoS, resource abuse

Compliance & Governance

Interaction	Audit trails, user consent evidence
Model	Performance accountability, SLAs
Security	Incident response, breach notification
Data	GDPR, data sovereignty proof
Operational	Cost governance, capacity planning

CSA AI Controls Matrix + NIST AI RMF

CSA AI Controls Matrix

Control Domains:

- Data Governance
- Model Security
- Operational Monitoring
- Incident Response

Key Controls:

- AIS-01: AI inventory and classification
- AIS-04: Model access logging
- AIS-07: Training data lineage
- AIS-12: Anomaly detection mechanisms

Telemetry Gap: Focuses on what to log, not how

NIST AI RMF

Core Functions:

Govern **Map** **Measure** **Manage**

Telemetry Implications:

Govern:
Policy compliance evidence

Map:
System context documentation

Measure:
Performance metrics collection

Manage:
Incident response data

Telemetry Gap: High-level guidance, implementation-agnostic

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